DOPE: DOWNSTREAM OBJECTIVE-PRESERVING ENCODING OF REGRESSION TABLES INTO KILOBYTE GENERATORS

Jepson Taylor[1], **Alton Alexander**[1]
[1]NEVERHUMAN Research

ABSTRACT

Synthetic tables ship as rows or as multi-megabyte neural generators, often scored by the metric that selected them. We introduce **D**ownstream **O**bjective-**P**reserving **E**ncoding (DOPE), which compiles a numeric regression table into a generator under a 10,240-byte release cap (median 1,895 bytes; $490/500$ fits within it): quantized marginals, a dependence model, a twelve-input neural residual for the target, and a noise model; sampling reads only the artifact. We score retention, a null-normalized train-on-synthetic/train-on-real ratio (Esteban et al., 2017), under three auditors on 100 PMLB regression tables with paired, family-clustered intervals. DOPE's profile was chosen at one fit seed; we refit at five. Under CatBoost at synthetic size $4n$, DOPE's median retention is 0.898 $[0.804, 0.972]$. DOPE cannot be separated from TabSyn (scaled schedule), the strongest full-panel generator (Hodges–Lehmann difference -0.003 $[-0.053, 0.053]$), whose $42.3\,\mathrm{MB}$ median artifact is 2.2×10^4 times DOPE's, and cannot be separated from author-default ARF (0.048 $[-0.034, 0.149]$). On a separate 6-lineage block, DOPE is below Forest-Flow (-0.231 $[-0.382, -0.005]$). It is above a predictor-only sampler (0.197 $[0.112, 0.267]$) and is below four-fold resampled real rows (-0.179 $[-0.276, -0.074]$). All estimates use a validation cut; the official test split is reserved for one pre-registered run.

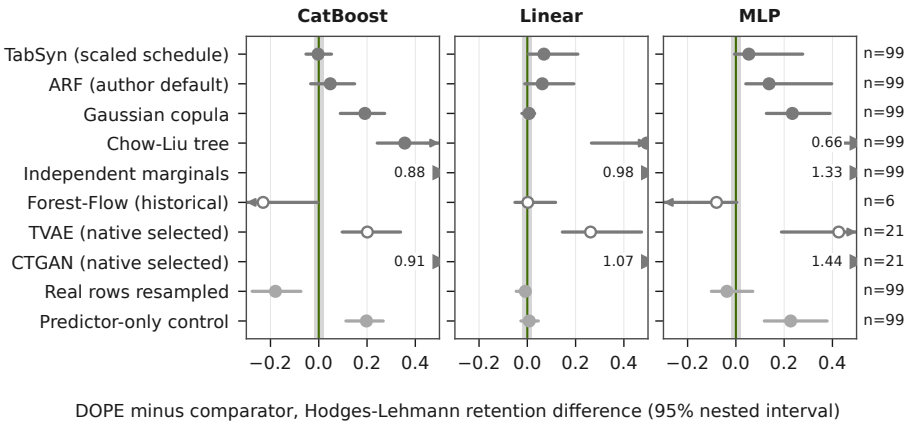


Figure 1: **DOPE minus the main comparators and both controls at synthetic size $4n$.** Hodges–Lehmann paired retention difference with its 95% nested bootstrap interval; the vertical line marks zero, and Table 1 adds native-selected ARF and the untuned default. Open marks are cohorts below 90 lineages, grey marks are controls, and the number beside a row is its CatBoost paired lineage count. The band is the equivalence margin ± 0.02; the axis runs from -0.3 to 0.5, with an edge arrow and its value for any difference beyond it. Holm families are in Table 1.

1 INTRODUCTION

A synthetic table is useful when a model trained on it behaves like a model trained on the real table. Two practices make that property hard to compare across generators. First, released generators are large: CTGAN and TVAE (Xu et al., 2019), TabDDPM (Kotelnikov et al., 2023), TabSyn (Zhang et al., 2024), and forest flows (Jolicoeur-Martineau et al., 2024) ship hundreds of kilobytes to gigabytes of weights, which are expensive to audit, to store next to a dataset, and to reason about as a release. Second, a generator paper often reports the metric that also chose its configuration, so selection and measurement are entangled.

DOPE starts from the downstream task. When the intended use is regression on a target y, a generator only has to reproduce the feature distribution well enough and the conditional mean $\mathbb{E}[y \mid x]$ well enough for a learner to recover it. DOPE therefore stores the conditional mean explicitly, as a tiny residual network over a screened set of inputs, and stores the features as quantized marginals tied together by a compact dependence model. The whole artifact must fit a byte budget, and sampling reads nothing but the artifact.

We evaluate with one measurement for every method. Retention (3) asks how much of a real-trained auditor's improvement over a null an auditor recovers when trained on synthetic rows. Two controls separate the ingredients: resampling real rows four-fold isolates the effect of row count, and a predictor-only sampler, which keeps a fitted regressor but drops the joint feature model, isolates the dependence model. Every comparison is paired by lineage, uses five DOPE fit seeds, and carries a nested bootstrap interval with simulated source families as clusters. Each method's selection data are stated.

Contributions.

1. **A kilobyte generator compiled from the table.** DOPE stores per-feature marginals and transforms, a dependence model chosen among seven families on the training rows, a twelve-input residual for the target, and a heteroscedastic noise model, all quantized into an artifact whose median charge is $1{,}895$ bytes (Section 3, Algorithm 1).
2. **A selection-aware, controlled measurement.** Retention is credited as a TSTR/TRTR ratio and computed by three auditors at sizes n and $4n$, beside a row-count control and a predictor-only control, with pre-declared Holm families and a robust equivalence rule (Section 4).
3. **Where DOPE stands, and why.** Under CatBoost at $4n$, DOPE cannot be separated from TabSyn (scaled schedule) while storing 2.2×10^4 times fewer bytes, cannot be separated from author-default ARF, and is above the predictor-only control; resampled real rows remain the upper reference (Section 5).
4. **A release gate that fails closed.** MFS-v3 is a gated geometric mean with an executable conformance test. It is null for every method here, and the paper names the gates that keep it null (Section 3.3).

2 RELATED WORK

Generators. Copula and Bayesian-network models (Patki et al., 2016; Chow & Liu, 1968; Aas et al., 2009; Zhang et al., 2014) fit explicit, often small, joint models; marginal-based private synthesizers (McKenna et al., 2021; 2022; Cai et al., 2021) add formal privacy. GANs and VAEs (Xu et al., 2019; Park et al., 2018; Zhao et al., 2021), diffusion and score models (Kotelnikov et al., 2023; Zhang et al., 2024; Kim et al., 2023; Lee et al., 2023; Shi et al., 2025), forest flows (Watson et al., 2023; Jolicoeur-Martineau et al., 2024), language-model generators (Borisov et al., 2023; Solatorio & Dupriez, 2023), sequential CART (Nowok et al., 2016), and prior-fitted networks (Ma et al., 2023; Hollmann et al., 2025) optimize a likelihood, a denoising objective, or adversarial agreement on the joint table. DOPE optimizes a different thing: the target's conditional mean, explicitly, under a byte budget, with the joint feature model kept compact.

Compact and program models of tables. Bayesian synthesis of probabilistic programs (Saad et al., 2019) and CrossCat (Mansinghka et al., 2016) also produce small, inspectable generative models of tables. They target the joint distribution and posterior inference; DOPE targets a downstream regression objective and charges every artifact in bytes against a release cap.

Algorithm 1 DOPE compile and sample. Training reads the fit rows only; sampling reads the artifact only.

Compile(D_{fit}, B):

1. Fit each feature's marginal and transform T_j and record its missing rate m_j.
2. Keep the twelve features with the largest absolute rank correlation with y; call them S.
3. Train (1) with AdamW (learning rate 0.002, clipped gradients) for the profile's step budget; freeze (b, W) and refit (a, β, v) in closed form.
4. Fit the noise model on the residuals $y - \hat{y}(x)$.
5. For each dependence family, fit it on the Gaussianized features, quantize, encode, and record a proxy score and the gate shortfall.
6. Return the candidate with the smallest gate shortfall, then the fewest bytes; an artifact above the budget B fails the release gate.

Sample(A, N): for each of N rows, draw z from the stored dependence; set $x_j = T_j^{-1}\big(F_j^{-1}(\Phi(z_j))\big)$; set $y = \text{clamp}_{[0,1]}\big(\hat{y}(x_S) + \sigma(x)\,\varepsilon\big)$ with $\varepsilon \sim \mathcal{N}(0,1)$; replace each feature by a missing mark with probability m_j.

Evaluation. Train-on-synthetic/test-on-real comparisons originate with Esteban et al. (2017); utility and fidelity measures (Snoke et al., 2018; Raab et al., 2018; Alaa et al., 2022; Zhang et al., 2020) and benchmark suites (Qian et al., 2023; McElfresh et al., 2023; Erickson et al., 2026) broaden them. Membership attacks on synthetic data (Shokri et al., 2017; Hayes et al., 2018; van Breugel et al., 2023; Stadler et al., 2022) and holdout-referenced distances (Platzer & Reutterer, 2021) assess privacy empirically, which is distinct from differential privacy (Dwork & Roth, 2014). Our retention is a null-normalized TSTR/TRTR ratio; the new elements are the controls, the paired multi-seed design, and the byte axis.

3 DOPE

3.1 The artifact

A DOPE artifact for a numeric table with features $x \in \mathbb{R}^d$ and target y stores five parts. (i) For every feature, a quantized marginal, a monotone transform onto the unit interval, and a missing-value rate. (ii) A dependence model over the Gaussianized features, chosen among seven families: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, and a two-component mixture. (iii) A residual model for the target on a screened support x_S of at most twelve inputs, ranked by absolute rank correlation with y:

$$h_j = \tanh\big(b_j + w_j^\top x_S\big)\,, \qquad \hat{y}(x) = a + \beta^\top x_S + v^\top h, \tag{1}$$

with sixteen hidden units. (iv) A heteroscedastic noise model: a linear fit of $\log|y - \hat{y}|$ on up to four screened features, with its scale clamped between the lower and upper deciles of the absolute residuals. (v) The positions of the screened columns. The compiler quantizes every part and encodes the artifact; the byte charge is the encoded model file plus the `projection.json` that maps raw columns onto the unit interval. An artifact larger than 10,240 bytes fails the release gate.

3.2 Compile, sample, and profile

Algorithm 1 lists the two procedures. The compile search reads the fit rows only. The reported profile, `features12_steps2048`, trains for 2,048 steps; the untuned trainer default trains for 96 steps with a width between 4 and 12. The reported profile was chosen on a six-lineage discovery panel and checked on six disjoint confirmation lineages, by validation retention at fit seed 11; those twelve lineages belong to the panel below. Section 5.6 and Tables 1 and 2 report the untuned default beside it, and the sealed test removes this selection from the final estimate.

3.3 Release gate

Retention alone cannot decide a release: an artifact can score well and still copy rows, miss the joint distribution, or exceed the byte cap. MFS-v3 combines seven components $c_i \in [0, 1]$ (utility transfer

0.35; representation closeness, driver fidelity, and distribution fidelity 0.15 each; structure fidelity 0.10; coverage realism and compactness 0.05 each) as

$$\text{MFS-v3} = \text{clip}_{[0,100]}\left(100\exp\left(\frac{\sum_i w_i \ln(\max(c_i,\epsilon))}{\sum_i w_i}\right)\right), \quad \epsilon = 10^{-6}, \tag{2}$$

and returns null unless four prerequisites hold (a real-versus-real control, both size cells, artifact-only sampling, locked metric implementations) and every hard gate passes; membership, attribute, and copy failures are gates, so utility cannot offset them. Compactness is $\text{clamp}(1-\text{bytes}/10240, 0, 1)$. A conformance test ties (2) to the contract and the scorer (supplement B). The score is null for every method in this paper because its privacy and representation gates are still unmeasured.

4 Protocol

4.1 Data and partitions

The panel holds the 100 rights-cleared PMLB regression lineages (Olson et al., 2017; Romano et al., 2022) that passed the data lock; 65 of them come from four simulated families (Feynman, Strogatz, Friedman-style `fri`, and BNG generators). Large sources are reduced by a fixed coreset to at most 800 fit rows. The official training rows are split 80/20 by seed 1729 into fit and validation rows, grouping duplicates. The official test rows stay unread until a single pre-registered run; every number here is a validation estimate.

4.2 Retention

Let L_{null}, L_{real}, and L_{syn} be an auditor's validation mean squared error, on the target min–max projected to $[0,1]$ by the training split, when it predicts the training mean, when it is trained on the real fit rows, and when it is trained on synthetic rows. Retention is

$$R = \frac{L_{\text{null}} - L_{\text{syn}}}{L_{\text{null}} - L_{\text{real}}}, \tag{3}$$

so $R = 1$ matches the real-trained auditor and $R < 0$ is worse than the null. A lineage is informative for an auditor when $L_{\text{null}} - L_{\text{real}} \geq 0.01\,|L_{\text{null}}|$. The auditors are CatBoost (Prokhorenkova et al., 2018), ordinary least squares, and a one-hidden-layer MLP; CatBoost and the MLP use seed 1729. A synthetic table has n or $4n$ rows, where n is the fit-row count, and three sample seeds. A fit's value is the median over sample seeds; a lineage's value is the median over its fits.

4.3 Comparators and controls

Comparators are author-default and native-selected ARF (Watson et al., 2023) at five fit seeds; TabSyn with a scaled schedule of 200 VAE and at most 1,000 diffusion epochs, which the authors' early stop can end sooner, at five fit seeds; Gaussian copula, Chow–Liu, and independent marginals selected by held-out log density; CTGAN and TVAE selected on validation skill of the linear and MLP auditors; and a historical Forest-Flow block. Supplement C lists each method's selection rule, data, and budget. The *predictor-only* control fits a ridge regression on the fit rows, samples each feature independently from its marginal, and adds Gaussian residual noise; it keeps a regressor and drops the joint model. The *real-row* control resamples the fit rows with replacement to $4n$ rows; it measures what extra rows alone do and is an upper reference rather than a generator.

4.4 Statistics

Comparisons pair lineages that both arms complete (a five-seed arm needs three complete fits). The paired estimand is the Hodges–Lehmann pseudo-median of the lineage differences, the location the Wilcoxon signed-rank test addresses. Intervals come from a nested bootstrap with 10,000 draws: clusters (each simulated family is one cluster; every other lineage is its own), then lineages inside a drawn cluster, then fit seeds inside a drawn lineage, separately for each arm. Wilcoxon p-values are Holm-adjusted inside the pre-declared families of each size (Table 1). The Wilcoxon test treats lineages as independent; each verdict reads the clustered interval. A 90% interval inside ± 0.02 would

Table 1: **DOPE minus each comparator at synthetic size $4n$.** Hodges–Lehmann paired difference in retention with its 95% nested bootstrap interval for each auditor; n, W/T/L (DOPE higher, equal, lower), and the Holm-adjusted p-value are for CatBoost. Holm families are pre-declared per size: the primary family has 12 tests (author-default ARF, TabSyn, TabDDPM, and Forest-Flow under the common budget, by three auditors), the density family 9, a deep-generator family 12 (author-default CTGAN and TVAE, TabDiff, GReaT), a secondary family 9 (native-selected ARF, synthpop, SMOTE), and the control family 6; untested members keep their slot. A dash marks a descriptive row outside every family; the tested author-default CTGAN and TVAE rows are in supplement E. Bytes are the median charged artifact bytes over each method's lineages with a charged fit, with that lineage count beneath; for TVAE and CTGAN it exceeds the scored n.

Comparator	n	CatBoost	Linear	MLP	W/T/L	Holm p	Bytes
TabSyn (scaled schedule)	99	-0.003	0.069	0.054	53/0/46	1.000	42.3 MB
		$[-0.053, 0.053]$	$[0.005, 0.210]$	$[-0.006, 0.277]$			(100)
ARF (author default)	99	0.048	0.062	0.137	60/0/39	0.229	1.4 MB
		$[-0.034, 0.149]$	$[-0.011, 0.193]$	$[0.041, 0.397]$			(100)
ARF (native selected)	99	0.083	0.083	0.191	66/0/33	0.004	1.6 MB
		$[0.006, 0.191]$	$[0.006, 0.217]$	$[0.060, 0.431]$			(100)
Gaussian copula	99	0.191	0.007	0.234	83/0/16	1.46×10^{-13}	5.5 KB
		$[0.089, 0.273]$	$[-0.023, 0.030]$	$[0.126, 0.390]$			(100)
Chow-Liu tree	99	0.356	0.495	0.658	91/0/8	5.59×10^{-15}	29.9 KB
		$[0.242, 0.728]$	$[0.266, 0.816]$	$[0.435, 1.166]$			(100)
Independent marginals	99	0.881	0.982	1.327	99/0/0	5.13×10^{-17}	2.5 KB
		$[0.790, 0.964]$	$[0.926, 1.008]$	$[1.143, 1.629]$			(100)
TVAE (native selected)	21	0.201	0.262	0.426	19/0/2	—	308.6 KB
		$[0.097, 0.339]$	$[0.145, 0.473]$	$[0.189, 1.231]$			(34)
CTGAN (native selected)	21	0.912	1.066	1.439	21/0/0	—	731.0 KB
		$[0.731, 1.298]$	$[0.812, 1.695]$	$[0.788, 2.272]$			(35)
Forest-Flow (historical)	6	-0.231	0.002	-0.080	0/0/6	—	325.8 MB
		$[-0.382, -0.005]$	$[-0.051, 0.117]$	$[-0.562, 0.004]$			(6)
DOPE, untuned default	99	0.134	-0.003	0.138	87/0/12	—	1.7 KB
		$[0.053, 0.208]$	$[-0.026, 0.005]$	$[0.052, 0.251]$			(100)
Predictor-only control	99	0.197	0.008	0.227	92/0/7	2.80×10^{-14}	—
		$[0.112, 0.267]$	$[-0.027, 0.046]$	$[0.118, 0.378]$			
Real rows resampled	99	-0.179	-0.008	-0.037	10/0/89	2.02×10^{-13}	—
		$[-0.276, -0.074]$	$[-0.046, 0.008]$	$[-0.101, 0.070]$			

establish equivalence. No generator comparison meets that rule. These choices were committed before the five-seed refit ran, but after an earlier four-seed run whose samples the refit reproduces at seeds 23 to 71 (Section 5.6).

5 Results

5.1 DOPE against the strongest comparators

Table 1 and Fig. 1 give the paired differences at $4n$. The strongest full-panel comparator under CatBoost is TabSyn (scaled schedule): DOPE cannot be separated from it, with difference -0.003 (-0.053 to 0.053; $n = 99$; W/T/L 53/0/46). Against author-default ARF the CatBoost difference is 0.048 (-0.034 to 0.149; $n = 99$), against the Gaussian copula it is 0.191, and against Chow–Liu 0.356. Under the linear auditor the difference from the copula is 0.007 (-0.023 to 0.030) and from TabSyn (scaled schedule) it is 0.069 (0.005 to 0.210). No generator comparison is equivalent under the pre-declared rule; "cannot be separated from" means that the interval contains zero, not that the two methods are interchangeable. TVAE ($n = 21$) and CTGAN ($n = 21$) are small cohorts drawn with open marks; their tested author-default configurations are in supplement E. On the 6 lineages of the historical Forest-Flow block, DOPE is below Forest-Flow (-0.231, -0.382 to -0.005; W/T/L 0/0/6), whose median artifact is 325.8 MB; supplement E gives every row.

5.2 What the controls show

The predictor-only control keeps a fitted ridge regressor and drops the joint feature model, so its gap bundles DOPE's dependence model with DOPE's nonlinear residual. Under CatBoost, DOPE minus predictor-only is 0.197 (0.112 to 0.267); under the MLP it is 0.227 and under the linear auditor 0.008. A tree or network auditor trained on independently drawn features sees feature combinations

Table 2: **Retention level by method, auditor, and synthetic size.** Median over lineages with the 95% nested bootstrap interval beneath it; n is the CatBoost lineage count of that arm, and the linear and MLP counts, which drop lineages where that auditor is uninformative, are in supplement E. Five-seed arms use the median of at least three complete fits per lineage. Controls are references, not generators. No Holm family.

		CatBoost		Linear		MLP	
Method	n	n rows	$4n$ rows	n rows	$4n$ rows	n rows	$4n$ rows
DOPE (ours)	99	0.824 [0.73, 0.94]	0.898 [0.80, 0.97]	0.986 [0.96, 0.99]	0.991 [0.97, 1.00]	0.969 [0.90, 0.99]	1.133 [1.03, 1.25]
DOPE, untuned default	99	0.636 [0.39, 0.82]	0.704 [0.42, 0.88]	0.986 [0.97, 0.99]	0.993 [0.98, 1.00]	0.887 [0.80, 0.96]	0.991 [0.96, 1.06]
TabSyn (scaled schedule)	99	0.863 [0.75, 0.93]	0.924 [0.81, 0.96]	0.932 [0.79, 0.98]	0.952 [0.83, 0.99]	0.884 [0.73, 0.98]	1.013 [0.95, 1.14]
ARF (author default)	99	0.718 [0.57, 0.84]	0.822 [0.69, 0.89]	0.903 [0.79, 0.96]	0.938 [0.85, 0.98]	0.828 [0.63, 0.89]	0.991 [0.89, 1.09]
ARF (native selected)	99	0.671 [0.55, 0.83]	0.804 [0.65, 0.89]	0.894 [0.74, 0.97]	0.920 [0.83, 0.98]	0.805 [0.58, 0.90]	0.966 [0.84, 1.03]
Gaussian copula	99	0.610 [0.31, 0.78]	0.669 [0.37, 0.87]	0.961 [0.91, 0.99]	0.987 [0.96, 1.00]	0.800 [0.62, 0.89]	0.978 [0.89, 1.07]
Chow-Liu tree	99	0.466 [0.25, 0.58]	0.576 [0.29, 0.64]	0.636 [0.27, 0.76]	0.659 [0.28, 0.76]	0.454 [0.18, 0.63]	0.660 [0.35, 0.80]
Independent marginals	99	−0.046 [−0.08, −0.03]	−0.012 [−0.04, 0.00]	−0.050 [−0.08, −0.01]	0.002 [−0.03, 0.02]	−0.215 [−0.48, −0.07]	−0.049 [−0.16, −0.02]
TVAE (native selected)	21	0.606 [0.41, 0.74]	0.674 [0.45, 0.78]	0.764 [0.55, 0.86]	0.796 [0.58, 0.88]	0.721 [0.25, 0.80]	0.812 [0.64, 0.88]
CTGAN (native selected)	21	−0.074 [−0.48, 0.12]	−0.080 [−0.28, 0.22]	−0.056 [−0.38, 0.12]	−0.033 [−0.45, 0.12]	−0.247 [−1.07, 0.10]	0.033 [−0.34, 0.33]
Forest-Flow (historical)	6	0.977 [0.81, 1.00]	1.011 [0.91, 1.04]	0.981 [0.75, 1.08]	0.985 [0.98, 1.00]	0.885 [0.58, 1.00]	1.054 [0.93, 2.73]
Predictor-only control	99	0.596 [0.34, 0.73]	0.631 [0.38, 0.78]	0.970 [0.92, 0.99]	0.981 [0.95, 1.00]	0.849 [0.73, 0.91]	0.947 [0.88, 1.02]
Real rows resampled	99	—	1.018 [1.00, 1.03]	—	0.998 [0.99, 1.00]	—	1.174 [1.05, 1.23]

that the real table never contains, while a linear auditor depends mostly on the regressor. The real-row control resamples real rows to $4n$; DOPE minus that control is -0.179 (-0.276 to -0.074) under CatBoost and -0.037 under the MLP, so DOPE is below resampled real rows under CatBoost. Extra rows lift an MLP auditor even when they are copies, which is why MLP retentions above one appear.

5.3 Both synthetic sizes

Table 2 reports retention levels at n and $4n$ for every arm; the two sizes are co-primary in the original pre-registration, the v2 contract places $4n$ in the headline, and $4n$ gives each auditor four times as many synthetic rows as the real fit view. Moving from n to $4n$ takes DOPE's CatBoost median from 0.824 to 0.898 and raises the CatBoost median of 6 of 6 full-panel comparators; the CatBoost ranking of the full-panel arms is the same at both sizes. Resampling the real rows to $4n$ gives 1.018 under CatBoost and 1.174 under the MLP, which is what row count alone does to each auditor.

5.4 Utility per byte

Fig. 2 places each method's CatBoost retention at $4n$ against its median charged bytes. DOPE's median artifact is 1,895 bytes, and $490/500$ fits are within 10,240 bytes; the 10 fits above it are every fit of 4544_GeographicalOriginalofMusic and 588_fri_c4_1000_100 (19,019 to 22,233 bytes). Without those 2 lineages, a sensitivity added after review, DOPE's CatBoost median at $4n$ is 0.898 (0.803 to 0.975; $n = 97$), and DOPE cannot be separated from TabSyn (scaled schedule) (-0.008, -0.060 to 0.050; $n = 97$). The strongest full-panel comparator's median artifact is 42.3 MB, and author-default ARF's is 1.4 MB. Over every measured generator, including DOPE's untuned default, the Pareto set of median bytes and CatBoost retention is {DOPE (ours); DOPE, untuned default; TabSyn (scaled schedule); Forest-Flow (historical)}. The points are unpaired medians that order methods without testing them, and a hollow point rests on fewer than 90 lineages; Table 1 gives the paired comparison. The median artifacts of the neural, diffusion, and forest generators are 1.6×10^2 to 1.7×10^5 times DOPE's.

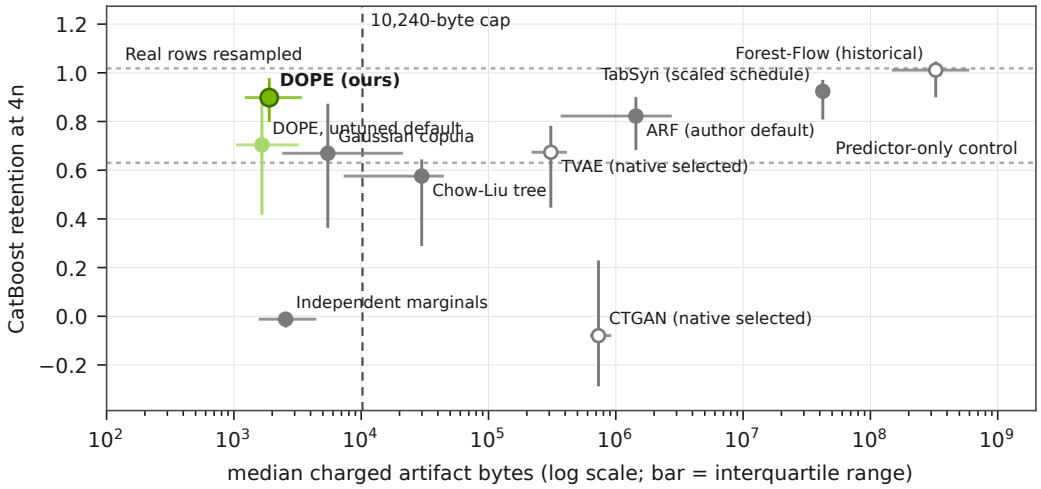


Figure 2: **CatBoost retention at $4n$ against median charged artifact bytes.** Points are lineage medians with 95% nested bootstrap intervals (vertical) and the interquartile range of bytes (horizontal). Hollow points are cohorts below 90 lineages. The light green point is DOPE's untuned default. Horizontal lines are the two controls, which store no artifact. The dashed line is the 10,240-byte release cap. Byte medians use each method's lineages with a charged fit, counted in Table 1; points come from unpaired cohorts, and Table 1 gives the paired comparison. No Holm family.

5.5 Fidelity and empirical privacy

Table 3 reports detection and distance measures at $4n$ on the fit-seed-11 samples of DOPE, ARF, the density baselines, TVAE, and CTGAN; TabSyn's distance and membership cells belong to the pre-declared privacy block. Forest-Flow's stored sample files support its retention re-score, but the privacy and detector runs record its cells as unavailable. A CatBoost two-sample detector, trained on at most 200 rows per class because the validation cut is that small, separates DOPE rows from validation rows with median AUC 0.640 (0.583 to 0.831), so the rows are distinguishable to a strong classifier even where an auditor's retention is high. The median distance from a DOPE row to its nearest fit row is 0.080 and to its nearest validation row 0.109; the fit view holds about four times as many rows as the validation cut, so the first distance is smaller even for a fresh draw. A distance-based membership attack reaches median AUC 0.505 (0.492 to 0.516). On the five-seed cells, DOPE's mean marginal Kolmogorov–Smirnov distance is 0.117 and its pairwise correlation fidelity 0.953, against 0.103 and 0.962 for author-default ARF and 0.126 and 0.958 for TabSyn. These are empirical measurements and give no guarantee of differential privacy or HIPAA de-identification. Supplement G gives the attack definitions and, for every method with scored samples including TabSyn, the CatBoost detector, marginal and pairwise fidelity, and α-precision and β-recall; Forest-Flow has marginal and pairwise fidelity only.

5.6 Seeds, profile selection, and sensitivity

The first version of this paper reported the historical fit-seed-11 population, with CatBoost median 0.940 at $4n$ when each sample seed also seeded its auditor; re-scored with auditor seed 1729, the same samples give 0.949. Four further seeds, run later with a newer binary, gave 0.862 on 99 lineages. The two binaries write byte-identical seed-11 artifacts on 100 of 100 lineages and byte-identical samples for seeds 23 to 71 on 400 of 400 fits, and the refit reproduces the historical seed-11 samples on 86 of 98 lineages. On the 97 lineages common to every cell, the historical binary gives 0.949 at seed 11 and 0.862 at seeds 23 to 71, and the later binary gives 0.949 and 0.862. Identical artifacts and samples leave the seed, not the binary, as the source of the gap. Seed 11 ranks first among the five seeds on 47 of 99 lineages, against 19.8 if the seeds were exchangeable. Because a seed acts on every lineage at once, as explained next, the lineages are not independent trials, so this count is descriptive and carries no p-value. Seeds 23 to 71 alone give 0.862 (0.771 to 0.958).

Table 3: **Detection and empirical privacy at $4n$, fit-seed-11 samples.** Medians over lineages with 95% family-cluster intervals; TabSyn's distance cells are pending, and the privacy run records Forest-Flow's cells as unavailable. C2ST is a CatBoost two-sample AUC (one half is indistinguishable). DCR fit and DCR val. are median distances to the nearest fit row and to the nearest validation row; the fit reference is about four times larger. NNDR is the nearest-neighbour distance ratio against the fit rows. MIA is a distance-attack AUC. n is the lineage count, given as a range where the metrics differ; a lineage counts when all three of its sample files resolve at the requested row count, a different rule from the informative-retention rule behind the n of Table 1. No Holm family.

Method	n	C2ST	DCR fit	DCR val.	NNDR	MIA AUC
DOPE (fit seed 11)	98	0.640 [0.58, 0.83]	0.080 [0.04, 0.13]	0.109 [0.05, 0.15]	0.905 [0.85, 0.94]	0.505 [0.49, 0.52]
ARF (author default)	100	0.612 [0.57, 0.69]	0.082 [0.03, 0.13]	0.109 [0.05, 0.16]	0.907 [0.86, 0.93]	0.556 [0.54, 0.58]
ARF (native selected)	100	0.609 [0.58, 0.71]	0.083 [0.04, 0.13]	0.111 [0.05, 0.16]	0.911 [0.86, 0.93]	0.541 [0.53, 0.56]
Gaussian copula	99–100	0.735 [0.69, 0.87]	0.086 [0.06, 0.13]	0.107 [0.08, 0.16]	0.920 [0.89, 0.96]	0.502 [0.49, 0.52]
Chow-Liu tree	99–100	0.706 [0.59, 0.94]	0.108 [0.09, 0.17]	0.129 [0.11, 0.19]	0.924 [0.89, 0.96]	0.520 [0.50, 0.54]
Independent marginals	100	0.858 [0.76, 0.97]	0.113 [0.09, 0.16]	0.132 [0.10, 0.18]	0.945 [0.91, 0.97]	0.508 [0.49, 0.52]
TVAE (native selected)	22	0.731 [0.70, 0.89]	0.080 [0.05, 0.15]	0.112 [0.09, 0.19]	0.906 [0.85, 0.93]	0.532 [0.49, 0.54]
CTGAN (native selected)	22	0.954 [0.85, 0.98]	0.153 [0.09, 0.19]	0.187 [0.12, 0.22]	0.941 [0.91, 0.96]	0.502 [0.47, 0.52]

A fit seed fixes the residual network's initial weights, which are the same on every lineage with the same number of screened inputs, so one seed acts on the whole panel at once; seed 11 is also the seed at which the profile was chosen. The five-seed median in the abstract is the estimate this paper stands on. The untuned trainer default gives 0.704 under CatBoost at $4n$, and the reported profile is above it (0.134, 0.053 to 0.208), a difference that includes whatever the profile selection added. Supplement E gives every diagnostic cell and repeats the source-family sensitivity on the five-seed panel: outside the four simulated families, DOPE cannot be separated from TabSyn (scaled schedule) under CatBoost at $4n$ (-0.013, -0.078 to 0.051; $n = 34$) and cannot be separated from author-default ARF (0.015, -0.069 to 0.081; $n = 34$). In the pre-declared all-five-fits sensitivity, which keeps a lineage only where every fit seed completed, DOPE cannot be separated from TabSyn (scaled schedule) (-0.008, -0.059 to 0.048; $n = 97$).

6 Limitations

Validation estimates. Validation rows informed DOPE's profile (by retention), the CTGAN and TVAE configurations (by auditor skill), and the native-selected density and ARF configurations (by held-out log density); the sealed test run is the only estimate free of that selection. **Row count and the regressor.** Retention at $4n$ mixes fidelity with row count, which the real-row control bounds, and the predictor-only gap credits DOPE's dependence model and its nonlinear residual together. **Scope.** The panel is regression only, fit views hold at most 800 rows, and 65 of 100 lineages are simulated; clustering by family widens the intervals but cannot add independent sources. **Baselines.** TabSyn runs a scaled schedule of 200 VAE and at most $1{,}000$ diffusion epochs against the authors' $4{,}000$ and $10{,}001$, TVAE and CTGAN cover small cohorts, and Forest-Flow is a historical block of 6 lineages. The pre-declared five-seed density runs, synthpop, SMOTE, Forest-Flow under the common budget, TabDDPM, TabDiff, and an LLM generator are pending. **Seeds.** The density baselines, CTGAN, and TVAE contribute one fit seed each, so their side of each interval omits fitting variance. A DOPE fit seed acts on every lineage at once, so the five seeds are five draws of a panel-wide effect, and the nested interval, which resamples seeds within each lineage, does not widen for that shared effect. **Privacy.** The privacy rows are empirical attacks on released rows; attacks on the artifact itself and shadow-model attacks are pending.

7 Conclusion

DOPE compiles a regression table into a generator whose median charge is $1{,}895$ bytes. Under CatBoost at $4n$ it retains 0.898 of a real-trained auditor's gain, cannot be separated from TabSyn (scaled schedule) at about 2.2×10^4 times fewer bytes, and is above a predictor-only sampler. The release score is null until its privacy and representation gates are measured, and the official test split waits for one pre-registered run. The analysis contract, every ledger, and the scripts that regenerate

each number from those ledgers are public, so each figure and table in this paper can be rebuilt from the repository with one command.

AVAILABILITY

The analysis contract (`predeclare_v2.json`) was committed before the five-seed refit ran, after an earlier four-seed run, and a continuous-integration job deletes every generated input and rebuilds the paper from the committed ledgers. Jankurai is the repository's conformance auditor. A passing audit means the build, tests, and boundary proofs cleared the release floor with no high finding. It is not a utility score, and it is separate from any MFS value and from any privacy certificate. The implementation and the receipts behind these tables are at

https://github.com/neverhuman/dope.

REFERENCES

Kjersti Aas, Claudia Czado, Arnoldo Frigessi, and Henrik Bakken. Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2):182–198, 2009. doi: 10.1016/j.insmatheco.2007.02.001.

Ahmed Alaa, Boris Van Breugel, Evgeny S. Saveliev, and Mihaela van der Schaar. How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 290–306. PMLR, 2022. doi: 10.48550/ARXIV.2102.08921. URL https://proceedings.mlr.press/v162/alaa22a.html.

Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. Language models are realistic tabular data generators. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.06280. URL https://openreview.net/forum?id=cEygmQNOeI.

Kuntai Cai, Xiaoyu Lei, Jianxin Wei, and Xiaokui Xiao. Data synthesis via differentially private markov random fields. *Proceedings of the VLDB Endowment*, 14(11):2190–2202, 2021. doi: 10.14778/3476249.3476272.

C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467, May 1968. doi: 10.1109/TIT.1968.1054142.

Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–487, 2014. doi: 10.1561/0400000042.

Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, David Salinas, and Frank Hutter. TabArena: A living benchmark for machine learning on tabular data. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, pp. 17285–17350, 2026. doi: 10.52202/085713-0519. URL https://openreview.net/forum?id=jZqCqpCLdU.

Cristóbal Esteban, Stephanie L. Hyland, and Gunnar Rätsch. Real-valued (medical) time series generation with recurrent conditional GANs. *arXiv preprint arXiv:1706.02633*, 2017. doi: 10.48550/ARXIV.1706.02633.

Jamie Hayes, Luca Melis, George Danezis, and Emiliano De Cristofaro. LOGAN: Membership inference attacks against generative models. *Proceedings on Privacy Enhancing Technologies*, 2019(1):133–152, 2018. doi: 10.2478/popets-2019-0008.

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.

Alexia Jolicoeur-Martineau, Kilian Fatras, and Tal Kachman. Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pp. 1288–1296. PMLR, 2024. doi: 10.48550/ARXIV.2309.09968. URL https://proceedings.mlr.press/v238/jolicoeur-martineau24a.html.

Jayoung Kim, Chaejeong Lee, and Noseong Park. STaSy: Score-based tabular data synthesis. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.04018. URL https://openreview.net/forum?id=1mNssCWt_v.

Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. TabDDPM: Modelling tabular data with diffusion models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 17564–17579, 2023. doi: 10.48550/ARXIV.2209.15421. URL https://proceedings.mlr.press/v202/kotelnikov23a.html.

Chaejeong Lee, Jayoung Kim, and Noseong Park. CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 18940–18956. PMLR, 2023. doi: 10.48550/ARXIV.2304.12654. URL https://proceedings.mlr.press/v202/lee23i.html.

Junwei Ma, Apoorv Dankar, George Stein, Guangwei Yu, and Anthony Caterini. TabPFGen – tabular data generation with TabPFN. In *NeurIPS 2023 Second Table Representation Learning Workshop*, 2023. URL https://openreview.net/forum?id=4MkkNsAEmO.

Vikash Mansinghka, Patrick Shafto, Eric Jonas, Cap Petschulat, Max Gasner, and Joshua B. Tenenbaum. CrossCat: A fully Bayesian nonparametric method for analyzing heterogeneous, high dimensional data. *Journal of Machine Learning Research*, 17(138):1–49, 2016. URL https://jmlr.org/papers/v17/11-392.html.

Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Vishak Prasad C, Ganesh Ramakrishnan, Micah Goldblum, and Colin White. When do neural nets outperform boosted trees on tabular data? In *Advances in Neural Information Processing Systems*, volume 36, pp. 76336–76369, 2023. doi: 10.52202/075280-3337. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/f06d5ebd4ff40b40dd97e30cee632123-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.

Ryan McKenna, Gerome Miklau, and Daniel Sheldon. Winning the NIST contest: A scalable and general approach to differentially private synthetic data. *Journal of Privacy and Confidentiality*, 11(3), 2021. doi: 10.29012/jpc.778.

Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. AIM: An adaptive and iterative mechanism for differentially private synthetic data. *Proceedings of the VLDB Endowment*, 15(11):2599–2612, 2022. doi: 10.14778/3551793.3551817.

Beata Nowok, Gillian M. Raab, and Chris Dibben. synthpop: Bespoke creation of synthetic data in R. *Journal of Statistical Software*, 74(11):1–26, 2016. doi: 10.18637/jss.v074.i11.

Randal S. Olson, William La Cava, Patryk Orzechowski, Ryan J. Urbanowicz, and Jason H. Moore. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining*, 10(1):36, 2017. doi: 10.1186/s13040-017-0154-4.

Noseong Park, Mahmoud Mohammadi, Kshitij Gorde, Sushil Jajodia, Hongkyu Park, and Youngmin Kim. Data synthesis based on generative adversarial networks. *Proceedings of the VLDB Endowment*, 11(10):1071–1083, 2018. doi: 10.14778/3231751.3231757.

Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. The synthetic data vault. In *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410, 2016. doi: 10.1109/DSAA.2016.49.

Michael Platzer and Thomas Reutterer. Holdout-based empirical assessment of mixed-type synthetic data. *Frontiers in Big Data*, 4:679939, 2021. doi: 10.3389/fdata.2021.679939.

Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018. doi: 10.48550/ARXIV.1706.09516. URL https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html.

Zhaozhi Qian, Rob Davis, and Mihaela van der Schaar. Synthcity: A benchmark framework for diverse use cases of tabular synthetic data. In *Advances in Neural Information Processing Systems*, volume 36, pp. 3173–3188, 2023. doi: 10.52202/075280-0140. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/09723c9f291f6056fd1885081859c186-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.

Gillian M. Raab, Beata Nowok, and Chris Dibben. Practical data synthesis for large samples. *Journal of Privacy and Confidentiality*, 7(3):67–97, 2018. doi: 10.29012/jpc.v7i3.407.

Joseph D. Romano, Trang T. Le, William La Cava, John T. Gregg, Daniel J. Goldberg, Praneel Chakraborty, Natasha L. Ray, Daniel Himmelstein, Weixuan Fu, and Jason H. Moore. PMLB v1.0: an open-source dataset collection for benchmarking machine learning methods. *Bioinformatics*, 38(3):878–880, 2022. doi: 10.1093/bioinformatics/btab727.

Feras A. Saad, Marco F. Cusumano-Towner, Ulrich Schaechtle, Martin C. Rinard, and Vikash K. Mansinghka. Bayesian synthesis of probabilistic programs for automatic data modeling. *Proceedings of the ACM on Programming Languages*, 3(POPL):1–32, 2019. doi: 10.1145/3290350.

Juntong Shi, Minkai Xu, Harper Hua, Hengrui Zhang, Stefano Ermon, and Jure Leskovec. TabDiff: a mixed-type diffusion model for tabular data generation. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2410.20626. URL https://openreview.net/forum?id=swvURjrt8z.

Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pp. 3–18, 2017. doi: 10.1109/sp.2017.41.

Joshua Snoke, Gillian M. Raab, Beata Nowok, Chris Dibben, and Aleksandra Slavkovic. General and specific utility measures for synthetic data. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 181(3):663–688, 2018. doi: 10.1111/rssa.12358.

Aivin V. Solatorio and Olivier Dupriez. REaLTabFormer: Generating realistic relational and tabular data using transformers, 2023.

Theresa Stadler, Bristena Oprisanu, and Carmela Troncoso. Synthetic data – anonymisation groundhog day. In *31st USENIX Security Symposium (USENIX Security 22)*, pp. 1451–1468, 2022. URL https://www.usenix.org/conference/usenixsecurity22/presentation/stadler.

Boris van Breugel, Hao Sun, Zhaozhi Qian, and Mihaela van der Schaar. Membership inference attacks against synthetic data through overfitting detection. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 3493–3514. PMLR, 2023. doi: 10.48550/ARXIV.2302.12580. URL https://proceedings.mlr.press/v206/breugel23a.html.

David S. Watson, Kristin Blesch, Jan Kapar, and Marvin N. Wright. Adversarial random forests for density estimation and generative modeling. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 5357–5375, 2023. doi: 10.48550/ARXIV.2205.09435. URL https://proceedings.mlr.press/v206/watson23a.html.

Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32, 2019. doi: 10.48550/ARXIV.1907.00503. URL https://papers.nips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html.

Hengrui Zhang, Jiani Zhang, Balasubramaniam Srinivasan, Zhengyuan Shen, Xiao Qin, Christos Faloutsos, Huzefa Rangwala, and George Karypis. Mixed-type tabular data synthesis with score-based diffusion in latent space. In *International Conference on Learning Representations*, 2024. doi: 10.48550/ARXIV.2310.09656. URL https://openreview.net/forum?id=4Ay23yeuz0.

Jun Zhang, Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and Xiaokui Xiao. PrivBayes: Private data release via bayesian networks. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*, pp. 1423–1434, 2014. doi: 10.1145/2588555.2588573.

Kevin Alex Zhang, Kalyan Veeramachaneni, and Carles Sala. SDMetrics: Synthetic data metrics. Open-source library, MIT Data To AI Lab, 2020. URL https://docs.sdv.dev/sdmetrics/.

Zilong Zhao, Aditya Kunar, Robert Birke, and Lydia Y. Chen. CTAB-GAN: Effective table data synthesizing. In *Proceedings of The 13th Asian Conference on Machine Learning*, volume 157 of *Proceedings of Machine Learning Research*, pp. 97–112. PMLR, 2021. URL https://proceedings.mlr.press/v157/zhao21a.html.